Approximate calculations with the derivative

Near one point a curve is a straight line — which is enough to compute √4.02 in your head.

LESSON 28–30 | 3 × 40 MIN | ALGEBRA 11, §2.1 | P1 · 9.4 · P2 · NUMERICAL METHODS

LESSON CLOCK

16' | 24' | 24' | 24' | 36' | 11

Where the formula comes from	16 min
Roots and powers	24 min
Changes in a quantity	24 min
Error estimates	24 min
Practice	36 min
Homework	11 min

LEARNING OBJECTIVES

- Derive and use the linear approximation $f(a + h) \approx f(a) + h \cdot f'(a)$.
- Estimate a root or a power without a calculator.
- Estimate the change in a quantity from a small change in another.
- Estimate the relative and percentage error in a computed quantity.

TEXTBOOK

National	pp. 131–146
Cambridge	Pure Mathematics 1, pp. 186–190
Workbook	P1 Exercise 9C

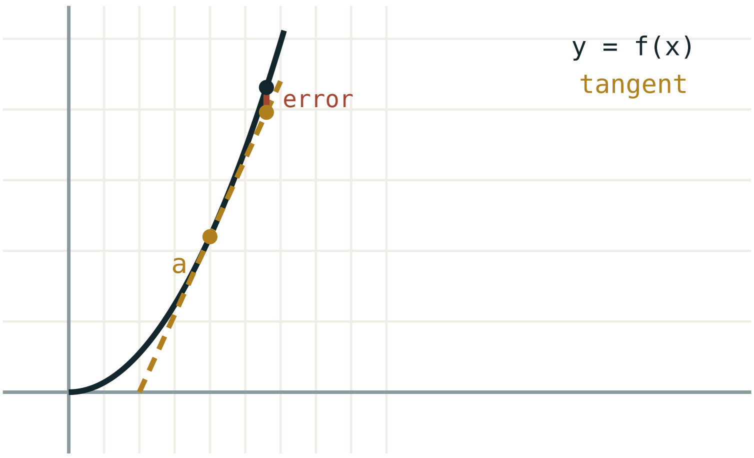
01 Explanation

Where the formula comes from

Return to the definition. For small h,

$$\frac{f(a+h) - f(a)}{h} \approx f'(a) \Rightarrow f(a+h) \approx f(a) + h \cdot f'(a)$$

Geometrically: near the point of tangency the curve and its tangent are almost the same line, so the tangent's height is a good estimate of the curve's.



near a the tangent and the curve agree

Close to a the two agree; the error grows with the square of the distance.

CHOOSING A

Pick the nearest point at which f and f' are **easy**. For $\sqrt{4.02}$ take $a = 4$; for $\sin 31°$ take $a = 30°$. The whole art is that choice.

Roots and powers

Estimate $\sqrt{4.02}$. Take $f(x) = \sqrt{x}$, $a = 4$, $h = 0.02$:

$$f(4) = 2,\ f'(x) = \frac{1}{2\sqrt{x}},\ f'(4) = 0.25$$

$$\sqrt{4.02} \approx 2 + 0.02 \times 0.25 = 2.005$$

The true value is $2.004994\ldots$ — the estimate is right to five decimal places, from two multiplications.

ESTIMATE	A	H	ANSWER
$\sqrt{9.06}$	9	0.06	3.01
$\sqrt{101}$	100	1	10.05
2.01^3	2	0.01	8.12
$\frac{1}{4.98}$	5	-0.02	0.2008

Changes in a quantity

The same formula, read as a statement about changes:

$$\Delta y \approx f'(x) \cdot \Delta x$$

Example. A cube's side is measured as 10 cm, with a possible error of 0.05 cm. By how much can the volume be out?

$$V = a^3,\ \frac{dV}{da} = 3a^2 = 300,\ \Delta V \approx 300 \times 0.05 = 15\ cm^3$$

So $V = 1000 \pm 15$ cm³. The absolute error in the side is small, but it is multiplied by $3a^2$ on its way to the volume.

Relative and percentage error

$$\text{relative error } \frac{\Delta y}{y} \approx \frac{f'(x) \cdot \Delta x}{f(x)}$$

For the cube: $\frac{15}{1000} = 1.5\%$, from a side error of only $\frac{0.05}{10} = 0.5\%$. The relative error is tripled, because the volume is the **third** power of the length.

A RULE WORTH REMEMBERING

For $y = x^n$ the relative errors satisfy $\frac{\Delta y}{y} \approx n \cdot \frac{\Delta x}{x}$. Squaring doubles the percentage error; cubing triples it; taking a square root halves it.

THE APPROXIMATION IS ONE-SIDED IN ACCURACY

It is good for **small** h only. Estimating $\sqrt{4.9}$ from $a = 4$ gives 2.1125 against a true 2.2136 — an error of 5%. Choose the nearer convenient point.

02 Worked examples

EXAMPLE 1 **Estimate $\sqrt{26}$ without a calculator.**

1. $a = 25, h = 1$

   $f(25) = 5$

2. $f'(x) = \dfrac{1}{2\sqrt{x}}, f'(25) = 0.1$

3. $\sqrt{26} \approx 5 + 1 \times 0.1 = 5.1$

   True: 5.0990.

ANSWER

≈ 5.1

EXAMPLE 2 **Estimate 3.02^4.**

1. $f(x) = x^4, a = 3, h = 0.02$

   $f(3) = 81$

2. $f'(x) = 4x^3, f'(3) = 108$

3. $81 + 0.02 \times 108 = 83.16$

   True: 83.17.

ANSWER

≈ 83.16

EXAMPLE 3 **A sphere's radius is 6 ± 0.1 cm. Estimate the error in its volume.**

1. $V = \dfrac{4}{3}\pi r^3$

   $\dfrac{dV}{dr} = 4\pi r^2 = 144\pi$

2. $\Delta V \approx 144\pi \times 0.1 \approx 45.2$

3. $V = 288\pi \approx 904.8$

4. Relative error $\dfrac{45.2}{904.8} = 5\%$

   Three times the $\dfrac{0.1}{6} \approx 1.67\%$ in r.

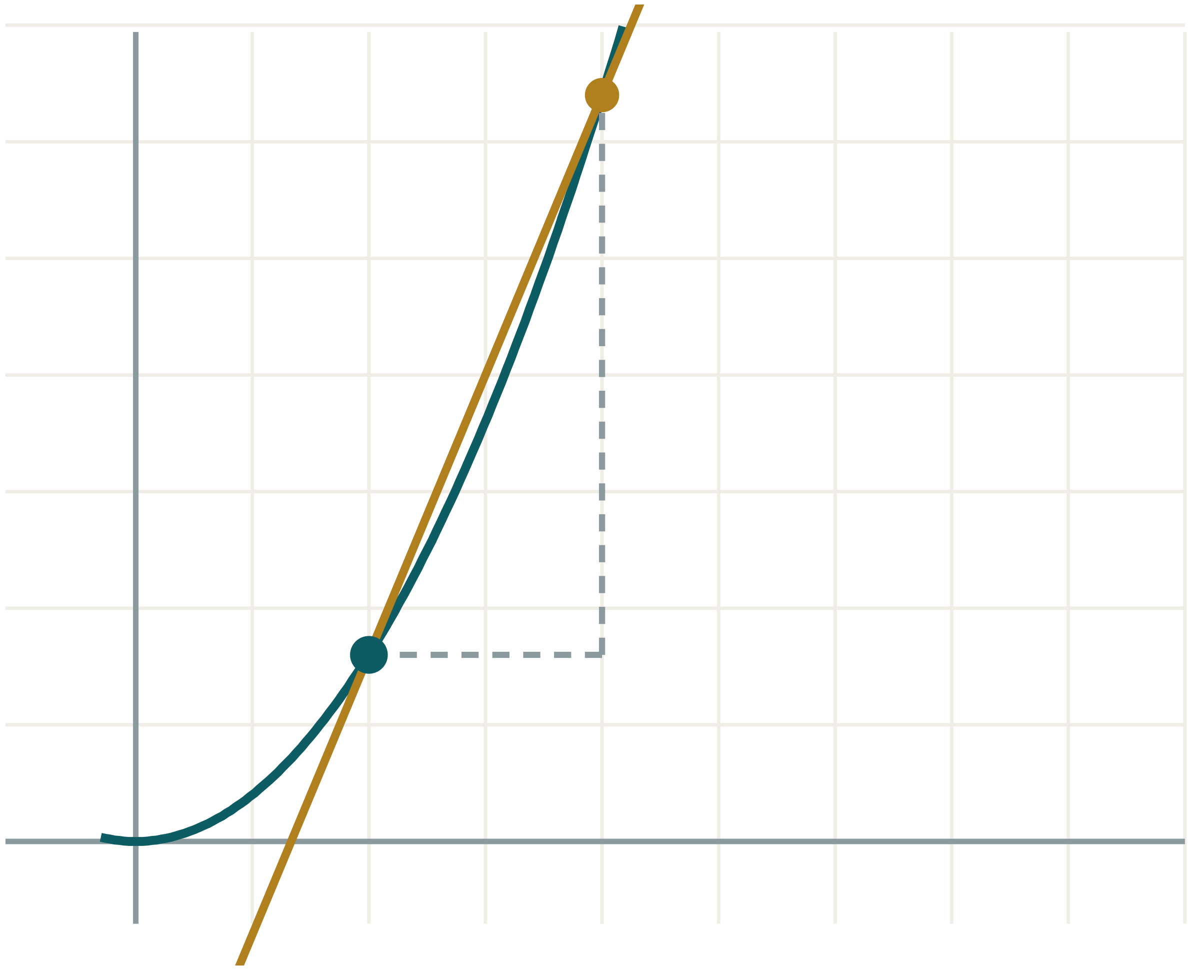
ANSWER

$\Delta V \approx 45$ cm³, about 5%

03 Interactive model

Shrink h and watch the tangent's prediction converge on the curve's value.

The tangent as a predictor



x **2** | h **2** | Δy / Δx **2.4** | f ′(x) exactly **1.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Approximation	Taqribiy hisoblash	Приближённое вычисление
Linear approximation	Chiziqli yaqinlashish	Линейное приближение
Small increment	Kichik orttirma	Малое приращение
Differential	Differensial	Дифференциал
Absolute error	Absolyut xatolik	Абсолютная погрешность
Relative error	Nisbiy xatolik	Относительная погрешность
Percentage error	Foizli xatolik	Процентная погрешность
Convenient point	Qulay nuqta	Удобная точка
Order of accuracy	Aniqlik tartibi	Порядок точности

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The linear approximation is:

- $f(a) \cdot h$
- $f(a) + h f'(a)$
- $f'(a) + h$
- $f(a + h)$

For $\sqrt{4.02}$ the best a is:

- 0
- 1
- 4
- 5

$\Delta y \approx$:

- $f'(x)\,\Delta x$
- $f(x)\,\Delta x$
- Δx
- $f'(x)$

For $y = x^3$ a 1% error in x gives about:

- 1%
- 2%
- 3%
- 9%

For $y = \sqrt{x}$ a 4% error in x gives about:

- 1%
- 2%
- 4%
- 8%

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Estimate $\sqrt{4.04}$

ANSWER ≈ 2.01

2. Estimate $\sqrt{9.06}$

ANSWER ≈ 3.01

3. Estimate $\sqrt{16.08}$

ANSWER ≈ 4.01

4. Estimate 2.01^2

ANSWER ≈ 4.04

5. Estimate 3.02^2

ANSWER ≈ 9.12

6. Estimate 1.02^3

ANSWER ≈ 1.06

7. Δy for $y = 5x$ when $\Delta x = 0.2$

ANSWER 1

Medium · the standard question

7 PROBLEMS

1. Estimate $\sqrt{26}$

   ANSWER ≈ 5.1

2. Estimate $\sqrt{101}$

   ANSWER ≈ 10.05

3. Estimate 3.02^4

   ANSWER ≈ 83.16

4. Estimate $\dfrac{1}{4.98}$

   ANSWER ≈ 0.2008

5. Cube side 10 ± 0.05; error in the volume

   ANSWER ≈ 15 cm³

6. Sphere radius 6 ± 0.1; error in the volume

   ANSWER ≈ 45 cm³

7. A 2% error in the radius gives what error in the area of a circle?

   ANSWER $\approx 4\%$

Hard · stretch and reason

7 PROBLEMS

1. Estimate $\sqrt[3]{8.12}$

ANSWER ≈ 2.01

2. Estimate $\sqrt{48}$ from $a = 49$

ANSWER ≈ 6.929 (true 6.928)

3. A circle's radius grows 1%. By what percentage does the area grow?

ANSWER $\approx 2\%$

4. A cube's surface area is measured to 2%. To what accuracy is the volume known?

ANSWER $\approx 3\%$

5. Estimate the error in $T = 2\pi\sqrt{\frac{L}{g}}$ from a 1% error in L

ANSWER $\approx 0.5\%$

6. Show the error in the linear approximation is of order h^2

ANSWER From the next term of the expansion

7. Why does estimating $\sqrt{4.9}$ from $a = 4$ fail badly?

ANSWER $h = 0.9$ is not small — the h^2 term dominates

07 Homework

Homework — 6 tasks

State a, h and $f'(a)$ before every estimate.

1. Estimate $\sqrt{36.6}$ and $\sqrt{99}$.
2. Estimate 2.03^5.
3. Estimate $\frac{1}{2.98}$.
4. A cube's side is 8 ± 0.02 cm. Find the volume and its absolute and percentage error.

5. A sphere's radius is measured to within 1%. To what accuracy is its volume known?
6. Explain in three sentences why the approximation is only good for small h.